



MULUNGUSHI UNIVERSITY

SCHOOL OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

NAME: TAONGA MUMBI

STUDENT NO: 202302522

PROGRAMMME: Bachelor of Science in Computer Science

COURSE: ICT 461 Web Systems

LECTURER: Mr. Nyirenda

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1. INTRODUCTION

The aim of this practical lab was to develop a simple course registration interface using standards-compliant HTML, responsive CSS and JavaScript. The practical also focused on accessibility, client-side form validation, browser Developer Tools and Git/GitHub as part of a professional development workflow.

2. OBJECTIVES

The main objective of this lab were;

- ♣ Create a semantic HTML course registration form.
- ♣ Use appropriate labels and input types for form controls.
- ♣ Add client-side validation and clear validation messages.
- ♣ Make the interface keyboard-friendly and responsive.
- ♣ Use browser Developer Tools to inspect the page and its network activity.
- ♣ Use Git to track the project and GitHub to store the project remotely.

3. Tools and technologies that were used in this lab

- ♣ HTML5
- ♣ CSS3
- ♣ JavaScript
- ♣ Visual Studio Code
- ♣ Web Browser Developer Tools
- ♣ Git
- ♣ GitHub

4. IMPLEMENTATION PROCESS

A course registration form was created containing fields for:

ICT461 Course Registration

ICT461 Registration

Web Systems and Technology

Full Name

Enter Name

Student ID

Enter student Id

Programme

-- Select --

Course

-- Select Course --

Register

- Full Name
- Student ID
- Programme
- Course

Each form control was associated with a visible label and required fields were marked appropriately. Validation was implemented using JavaScript to check the information entered by the user before displaying a successful registration message.

The page was also designed to be responsive so that the form remains usable on different screen sizes. A JavaScript submit event was used to prevent the default form submission and process the entered information.

5. TESTING AND ACCESSIBILITY

The form was tested using different types of input;

First, an empty form was submitted to verify that required fields were validated. Incorrect information was also entered into fields such as the Student ID to confirm that validation worked correctly.

The screenshot shows a web browser window with the title "ICT461 Course Registration". The address bar shows the file path "C:/Users/taonga%20Mumbi/Documents/ICT461-Lab%201/index.html". The main content area displays a registration form titled "ICT461 Registration" with the subtitle "Web Systems and Technology". The form contains four input fields: "Full Name" with the placeholder "Enter Name" and a red error message "Enter at least 4 letters and no numbers."; "Student ID" with the placeholder "Enter student Id" and a red error message "Student ID must be 6-10 uppercase letters/numbers."; "Programme" with a dropdown menu showing "-- Select --" and a red error message "Please select your programme."; and "Course" with a dropdown menu showing "-- Select Course --" and a red error message "Please select a course.". A blue "Register" button is at the bottom of the form. The Windows taskbar at the bottom shows the time as 8:38 PM on 9/16/2026.

FIGURE 2 shows an empty submitted form.

The screenshot shows the same registration form as Figure 2, but with incorrect data entered. The "Full Name" field contains "1234 jd" with a red error message "Enter at least 4 letters and no numbers.". The "Student ID" field contains "tao123" with a red error message "Student ID must be 6-10 uppercase letters/numbers.". The "Programme" dropdown menu still shows "-- Select --" with a red error message "Please select your programme.". The "Course" dropdown menu now shows "ICT461 - Web Systems". The blue "Register" button remains at the bottom. The Windows taskbar at the bottom shows the time as 8:39 PM on 9/16/2026.

FIGURE 3 shows incorrect details been entered.

Finally, valid information was entered and the registration success message was displayed.

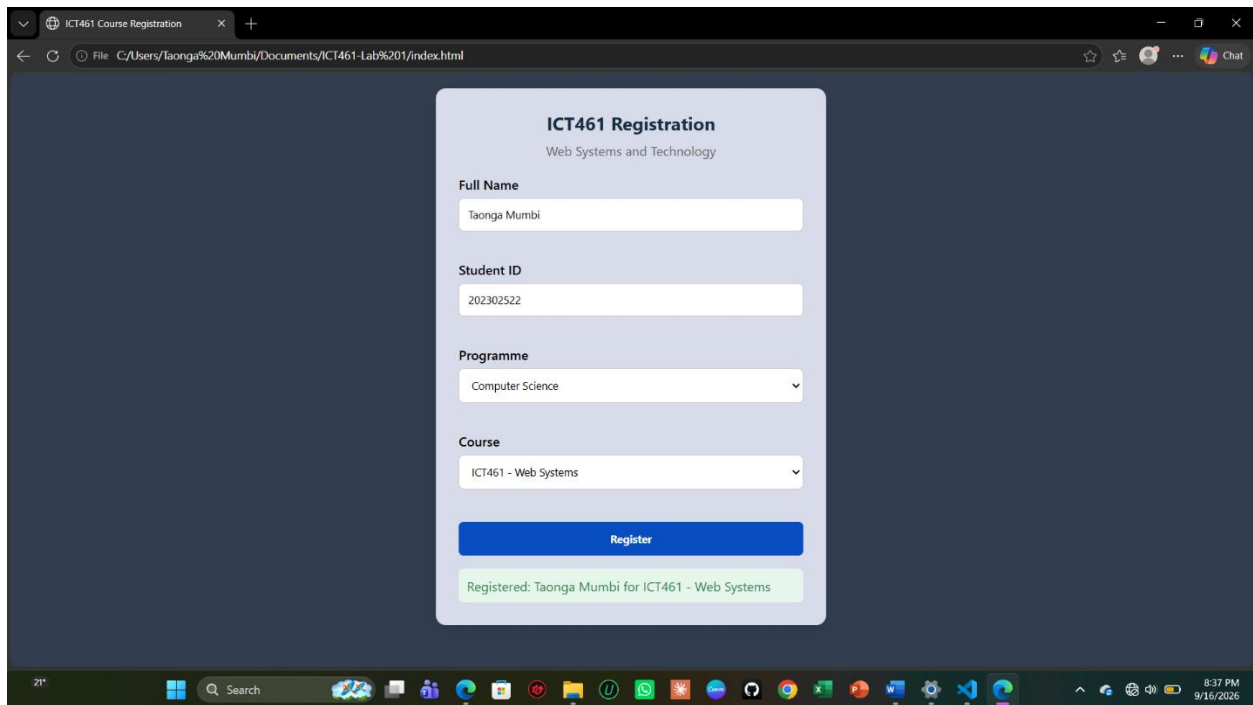


FIGURE 3 shows the successful course registration message.

Keyboard accessibility was also tested by navigating through the form using the Tab key and interacting with the controls without relying on the mouse.

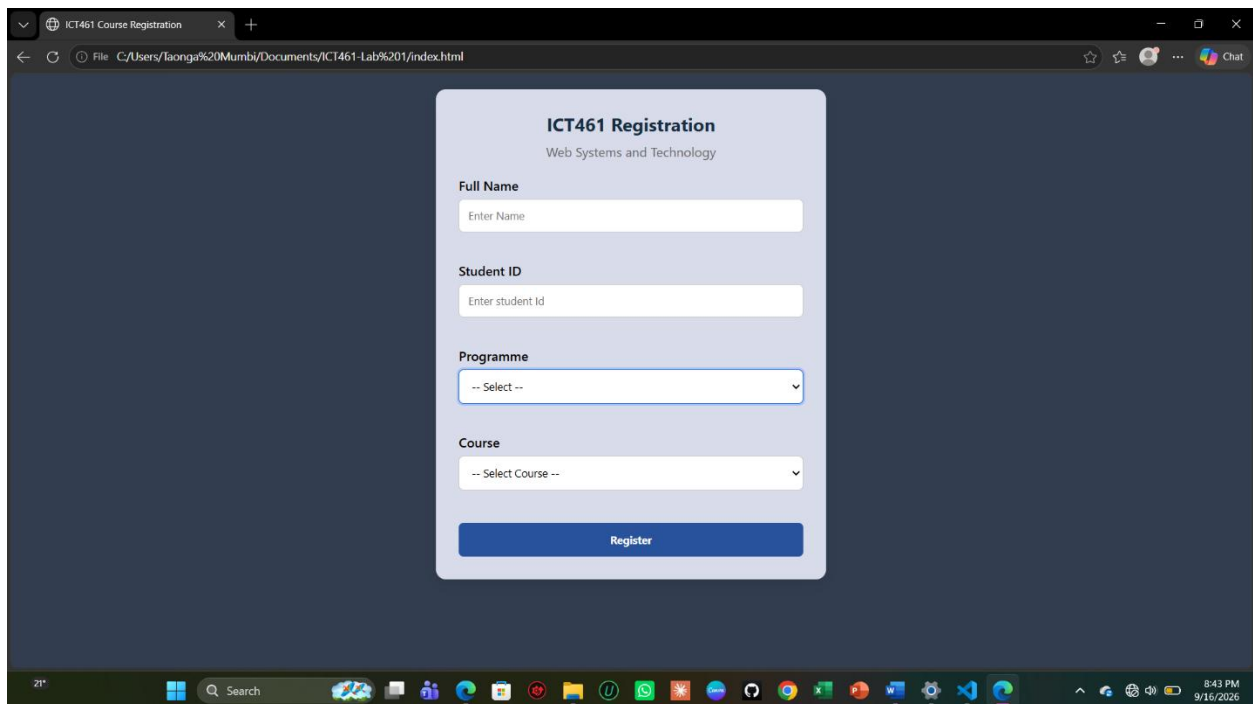


FIGURE 4 shows keyboard navigation through the registration form.

6. ELEMENT INSPECTION

The page was inspected using the browser's Developer Tools. The Elements panel was used to inspect the HTML structure and form elements.

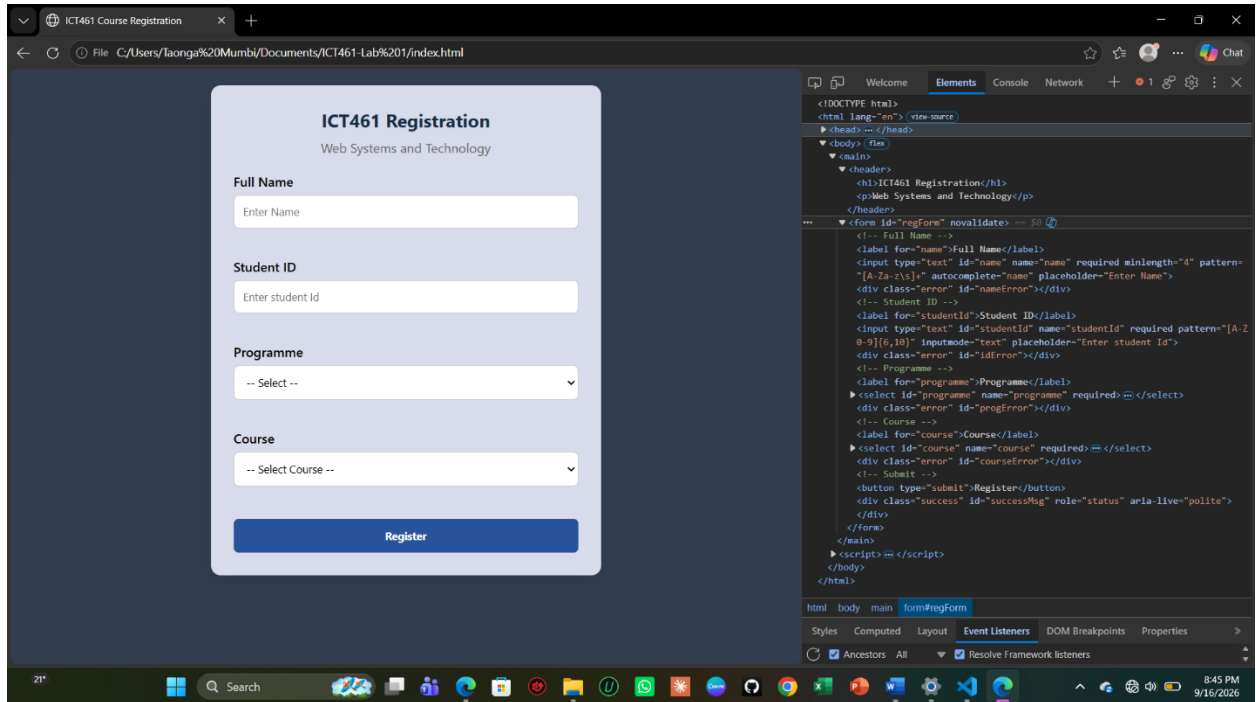


FIGURE 5 shows the elements panel showing the course registration form in HTML

7. NETWORK INSPECTION

The browser Network panel was used to observe the request made when the page was loaded. The request was inspected to identify information such as the request method, URL and response status.

The page was successfully loaded using a GET request with a successful 200 OK response.

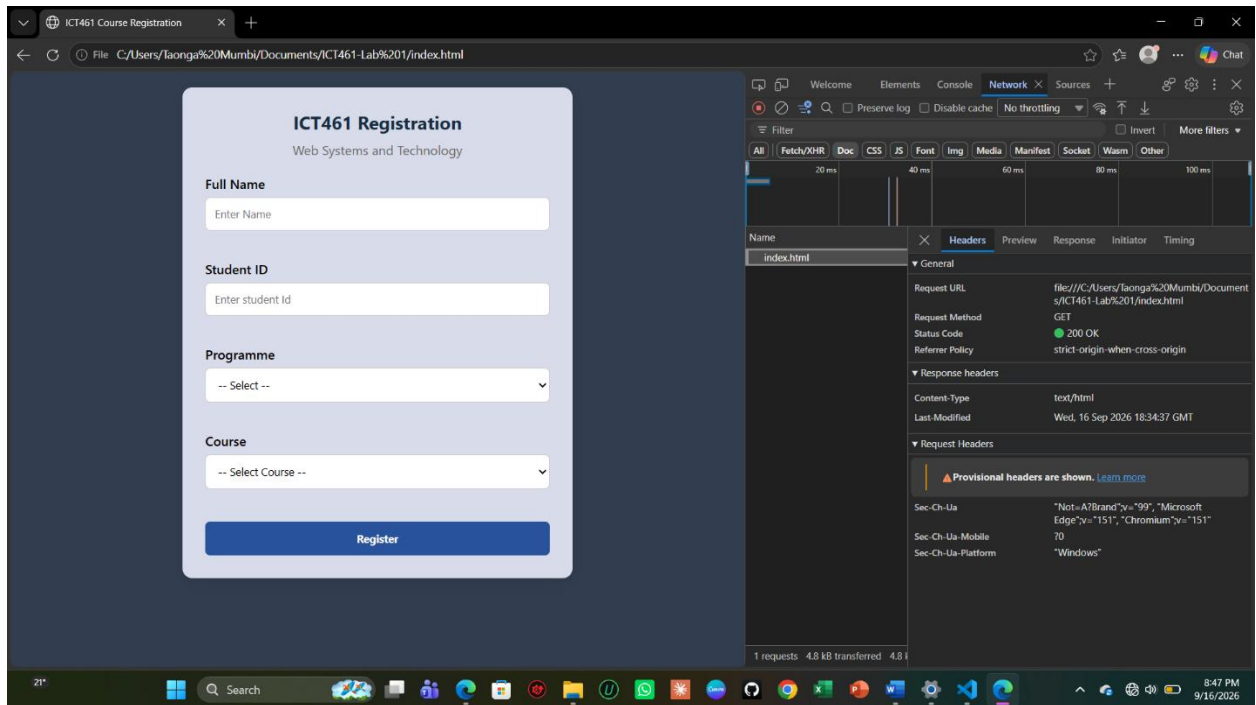


FIGURE 6 shows the page request of the network panel.

8. GIT and GITHUB

Git was initialized in the project directory and the project file were added to version control.

```

Problems  Output  Debug Console  Terminal  Ports
• PS C:\Users\Taonga Mumbi\Documents\ICT461-Lab 1> git init
  Initialized empty Git repository in C:/Users/Taonga Mumbi/Documents/ICT461-Lab 1/.git/
• PS C:\Users\Taonga Mumbi\Documents\ICT461-Lab 1> git status
  On branch master

  No commits yet

  Untracked files:
    (use "git add <file>..." to include in what will be committed)
      index.html
      script.js
      style.css

  nothing added to commit but untracked files present (use "git add" to track)

```

```
Problems  Output  Debug Console  Terminal  Ports

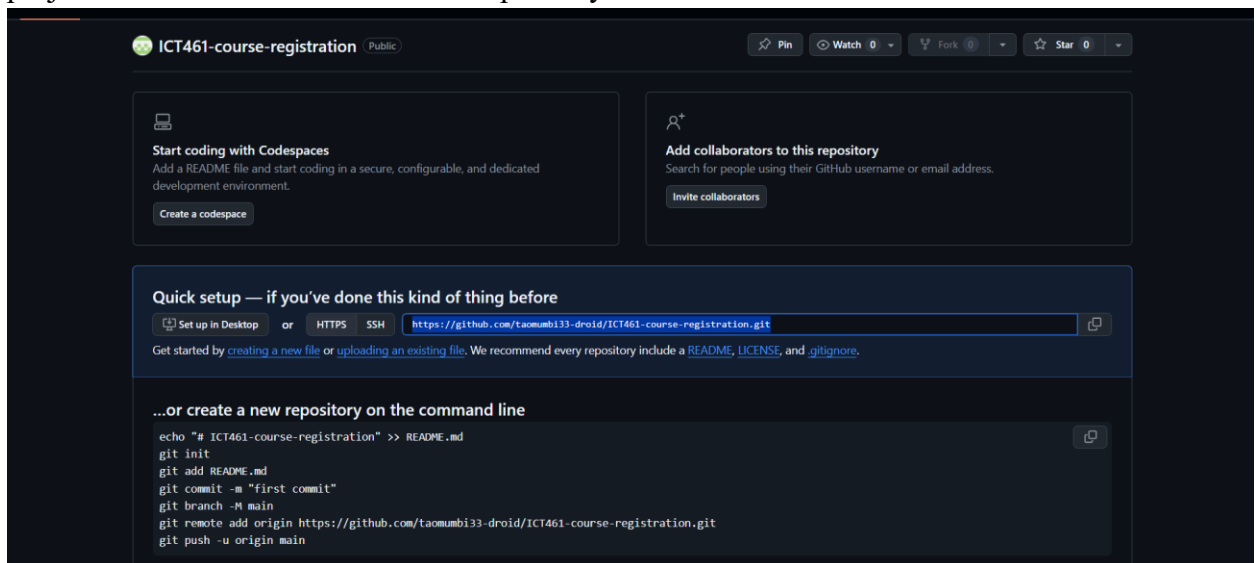
PS C:\Users\Taonga Mumbi\Documents\ICT461-Lab 1> git status
nothing added to commit but untracked files present (use "git add" to track)
PS C:\Users\Taonga Mumbi\Documents\ICT461-Lab 1>
PS C:\Users\Taonga Mumbi\Documents\ICT461-Lab 1> git add .
PS C:\Users\Taonga Mumbi\Documents\ICT461-Lab 1> git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   index.html

PS C:\Users\Taonga Mumbi\Documents\ICT461-Lab 1>
```

A GitHub repository named ICT461-course-registration was then created and the local project was connected to the remote repository.



The project was successfully pushed to GitHub, with the local master branch tracking origin/master.

GitHub Repository:

```
Problems  Output  Debug Console  Terminal  Ports

PS C:\Users\Taonga Mumbi\Documents\ICT461-Lab 1> git commit -m "create ICT461 course registration page"
create mode 100644 index.html
PS C:\Users\Taonga Mumbi\Documents\ICT461-Lab 1> git remote add origin https://github.com/taomumbi33-droid/ICT461-course-registration.git
PS C:\Users\Taonga Mumbi\Documents\ICT461-Lab 1> git push -u origin master
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 1.88 KiB | 1.88 MiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/taomumbi33-droid/ICT461-course-registration.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.
PS C:\Users\Taonga Mumbi\Documents\ICT461-Lab 1>
```

9. CONCLUSION

This practical helped me understand that developing a web page involves more than simply writing HTML and making the page look good. I learned how semantic HTML, labels, validation and keyboard accessibility contribute to making a web interface easier and safer to use. I also learned how JavaScript can be used to control form submission and provide feedback to the user without relying on the browser's default page refresh. Using Developer Tools was particularly useful because it allowed me to inspect the actual structure of my page and observe the network request made by the browser. This showed me that web development can be investigated using evidence from the browser rather than simply assuming that the code is working.

I also gained practical experience with Git and GitHub. I learned how to initialize a repository, stage files, create a commit, connect a local repository to a remote GitHub repository and push the project. This demonstrated the importance of version control when developing software.

One challenge I experienced was initially having the form refresh instead of displaying the expected result. Debugging the JavaScript and testing the form helped me understand the importance of preventing the browser's default form submission behaviour.

Overall, the practical improved my understanding of standards-compliant web development, accessibility, debugging and professional version-control practices.